AZ-305 · D4 — Infrastructure (Compute · Networking · Migrations)

WHERE it runs + HOW it connects · managed/PaaS minimizes admin · Global+HTTP=Front Door · Regional+HTTP=App Gateway · SQL MI subnet=/27

Compute service selection

I have a VM image / need OS control

- Single stateful workload → **Azure VM**
- Identical scale-out fleet, autoscale → **VMSS**
- Loose parallel: render, Monte Carlo → **Batch pool**
- MPI: CFD/weather/FEA + HB/HC + InfiniBand → **Batch MPI**
- 3rd-party scheduler → **CycleCloud**

I have a container

- Run one container fast, no orchestrator → **ACI**
- Scale-to-zero, per-request, HTTP/event, KEDA → **Container Apps**
- Full Kubernetes (mesh, controllers, multi-team) → **AKS**

I have code

- Web app / API, always-on → **App Service**
- Event-driven, ephemeral, scale-to-zero → **Functions**
- Workflow / SaaS connectors / no-code → **Logic Apps**

Scale-to-zero: ✅ ACA, Functions Consumption, AKS+KEDA · ❌ Functions Premium, App Service. **ACA = Linux only** → Windows web tier = App Service (Windows).

Load balancer selection

Global (multi-region) + HTTP/S, CDN, WAF	→ **Front Door**
Global + non-HTTP (TCP/UDP), DNS-based	→ **Traffic Manager**
Regional + HTTP/S, path/host, WAF, in-VNet	→ **App Gateway**
Regional + any TCP/UDP, ultra-low latency	→ **Load Balancer**

Global+HTTP=FD · Global+other=TM · Regional+HTTP=AppGW · Regional+other=LB

Front Door SKUs: Standard = CDN+LB+basic WAF · Premium = +advanced WAF (bot/custom) +Private Link origins.

TM routing: Priority = active/passive failover · Weighted = split / canary · Performance = lowest latency · Geographic = data sovereignty · Multivalue = many healthy IPs · Subnet = source-IP → endpoint. **TM cannot** rate-limit, SSL offload, or WAF.

VNet topology

Few VNets, single region, full control	→ **Hub-Spoke (self-managed hub)**
Many branches/regions, SD-WAN, min admin	→ **Virtual WAN (managed hub)**

vWAN = Microsoft-managed hub-spoke. Azure Firewall fits both: hub VNet (classic) or **Secured Virtual Hub** (vWAN).

Peering **non-transitive** → spoke-to-spoke routes via hub (UDR → Firewall/NVA next hop). **AMPLS:** 1 per workspace boundary; count private endpoints by VNet island — peered VNets share reach, isolated VNets need their own PE.

Connectivity options

≤1 Gbps, internet OK, encrypted IPsec	→ **Site-to-Site VPN**
Individual remote user laptop	→ **Point-to-Site VPN**
Private, 50 Mbps–100 Gbps, 99.95% SLA, low latency	→ **ExpressRoute**
Mission-critical resilience	→ **ER + VPN failover**
Many branches w/ SD-WAN	→ **Virtual WAN**
VNet ↔ VNet (Azure backbone)	→ **VNet peering**

ExpressRoute: Direct (no carrier, 10/100G) vs provider model · SKU Local < Standard < Premium (Premium = global).

Migration — "which R?"

Lift-and-shift to VM as-is	→ **Rehost**
Minor changes to use PaaS (SQL VM→SQL DB)	→ **Refactor**
Monolith→microservices, big restructuring	→ **Rearchitect**
Rewrite cloud-native from scratch	→ **Rebuild**
Adopt SaaS equivalent (CRM→Dynamics 365)	→ **Replace**
Keep on-prem (compliance)	→ **Retain**
Decommission, nobody needs it	→ **Retire**

Effort low→high: Retain < Rehost < Refactor < Rearchitect < Rebuild ≈ Replace.

CAF Migrate — 3 phases

Discover, inventory, rightsize, cost estimate	→ **Assess**
Replicate, migrate, cutover	→ **Deploy**
Stabilize, optimize, decommission on-prem	→ **Release**

Don't confuse with CAF lifecycle: **Strategy→Plan→Ready→Adopt→Govern→Manage**. Govern is NOT a migrate phase. Migrate sits in Adopt.

DDoS protection tier

Tier	When
Infrastructure (free)	Default everyone has
IP Protection	Small footprint (1-10 IPs), cost-sensitive
Network Protection	Enterprise, many IPs, DRR team, WAF discount

L3/L4 only. L7 (HTTP flood) → App Gateway WAF / Front Door WAF.

Data migration — which tool?

Whole server: DB + app + OS	→ **Azure Migrate**
SQL DB only → Azure SQL flavor	→ **Azure SQL Migration ext**
Assessment only, no execution	→ **DMA**
MySQL / PostgreSQL / MongoDB → Azure	→ **DMS**
Bulk files, online	→ **AzCopy / Storage Explorer**
Bulk files, offline	→ **Data Box / Import-Export**

Legacy SQL migration answer: DMS. **Online vs offline:** >~1 wk over the wire → ship offline.

Failure hierarchy & routing

Region → AZ → Fault Domain → Update Domain → Host

Availability Set = FD+UD in one datacenter (no AZ protection). **Availability Zones** = across AZs (datacenter loss). VMSS Flexible spans both.

🧠 Messaging — which service?

Pattern	Service
Pull, massive append stream (M/sec)	**Event Hubs**
One-to-one, FIFO/sessions/dead-letter	**Service Bus Queue**
One-to-many pub/sub, filtered future apps	**Service Bus Topic**
Push, react to discrete event	**Event Grid**